LOOSE COUPLING

Example 1: using @primary

```java
package com.example.LoosecouplingExample;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class LoosecouplingExampleApplication {
    public static void main(String[] args) {

        ApplicationContext context=SpringApplication.run(LoosecouplingExampleApplication.class, args);
        student s=context.getBean(student.class);
        s.show();
    }
}
```

```java
package com.example.LoosecouplingExample;

import org.springframework.context.annotation.Primary;
import org.springframework.stereotype.Component;

@Component
@Primary
public class pen implements writer {
    public void write()  1 usage
    {
        System.out.println("Student uses pen");
    }
}
```

```java
package com.example.LoosecouplingExample;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component  2 usages
public class student {
    @Autowired
    public student(writer writer) {
        this.writer = writer;
    }
    private writer writer;  2 usages

    public void show()  1 usage
    {
        System.out.println("Welcome");
        writer.write();
    }
}
```

```java
package com.example.LoosecouplingExample;

import org.springframework.stereotype.Component;

@Component
public class pencil implements writer {
    public void write()  no usages
    {
        System.out.println("Student uses pencil");
    }
}
```

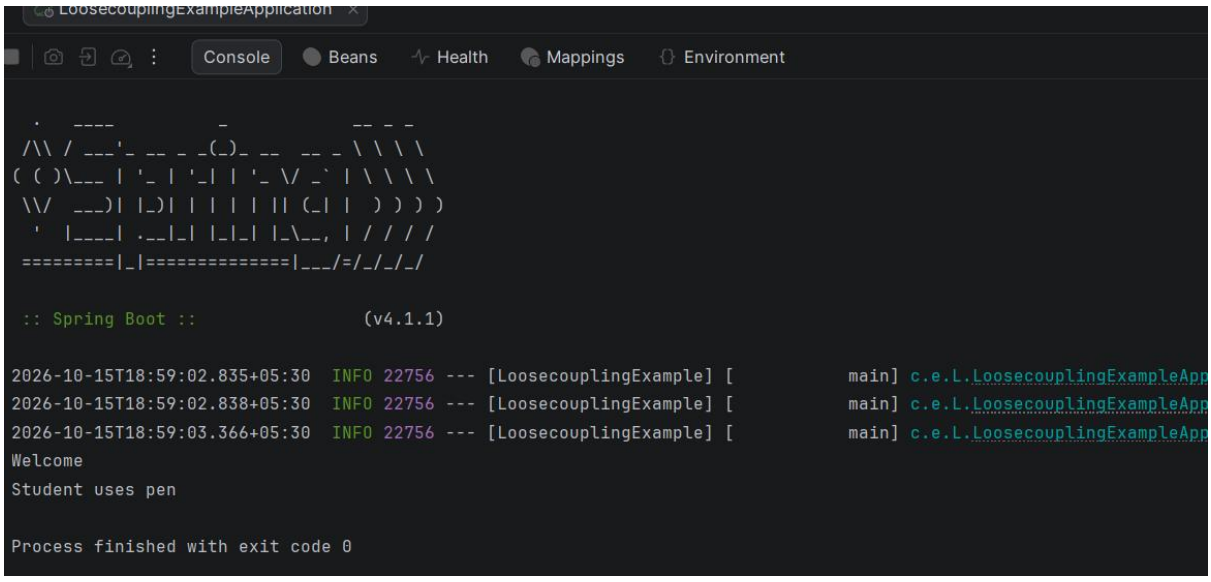
```java
package com.example.LoosecouplingExample;

import org.springframework.stereotype.Component;

@Component
public class sketch implements writer{
    public void write()  no usages
    {
        System.out.println("Student uses sketch");
    }
}
```

```java
package com.example.LoosecouplingExample;

public interface writer {  no usages  3 implementations
    public void write();  no usages  3 implementations
}
```



Example 2: Using qualifier

LoosecouplingApplication.java | Student.java | Mark.java | Average.java | Total.java | Calcul

```java
package com.example.Loosecoupling;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class LoosecouplingApplication {

    public static void main(String[] args) {
        ApplicationContext context=SpringApplication.run(LoosecouplingApplication.class, args);
        Student s=context.getBean(Student.class);
        s.display_details();
    }
}
```

```java
package com.example.Loosecoupling;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Qualifier;
import org.springframework.stereotype.Component;

@Component  2 usages
public class Student {
    int id=100;  1 usage
    String name="Suji";  1 usage

    @Autowired
    public Student( @Qualifier("total") Calculate calculate) {
        this.calculate = calculate;                //Usage of qualifier
    }

    private Calculate calculate;  2 usages

    public void display_details()  1 usage
    {
        System.out.println("ID: "+id);
        System.out.println("Name: "+name);
        calculate.display();
    }
}
```

```java
package com.example.Loosecoupling;

public interface Calculate {  5 usages  3 implementations
    public void display();  1 usage  3 implementations
}
```

```java
package com.example.Loosecoupling;

import org.springframework.context.annotation.Primary;
import org.springframework.stereotype.Component;

@Component
@Primary
public class Mark implements Calculate{
    @Override  1 usage
    public void display()
    {
        System.out.println("Mark Calculated");
    }
}
```

```java
package com.example.Loosecoupling;

import org.springframework.stereotype.Component;

@Component
public class Average implements Calculate{

    public void display()   1 usage
    {
        System.out.println("Average Calculated");
    }
}
```

```java
package com.example.Loosecoupling;

import org.springframework.stereotype.Component;

@Component
public class Total implements Calculate {

    public void display() {  1 usage
        System.out.println("Total calculated");
    }
}
```

```
2026-10-15T19:07:15.329+05:30  INFO 25548 --- [Loosecoupling] [           main] c.e.L.LoosecouplingApplication
2026-10-15T19:07:15.331+05:30  INFO 25548 --- [Loosecoupling] [           main] c.e.L.LoosecouplingApplication
2026-10-15T19:07:15.806+05:30  INFO 25548 --- [Loosecoupling] [           main] c.e.L.LoosecouplingApplication
ID: 100
Name: Suji
Total calculated

Process finished with exit code 0
```